

COMMON COURSE ASSIGNMENT

Cloud-Based College Event Registration System

A. Assignment Information

Particular	Details
Student Name	G.JAAHNAVI
Register Number	192525277
Department	AI&ML
Programme	B.Tech AIML
Course Code & Course Name	Cloud Computing
Academic Year / Batch	2025–2026
Faculty Name	Dr.K.Santhiya Lakshmi
Assignment Title	Cloud-Based College Event Registration System
Event Name	Coding Contest
Participation Type	Individual
Maximum Marks	100
Contact details	7386356934
Registration date	02-09-2026
Registration status	Confirmed

B. Assignment Problem / Challenge

Colleges regularly conduct events such as technical symposiums, cultural festivals, sports meets, workshops and hackathons, and these events attract registrations from a large number of students. Traditional registration methods, such as paper forms or manually maintained spreadsheets, make it difficult to consolidate entries, avoid duplicate or incomplete registrations, track participation category and generate quick, reliable statistics on turnout.

The challenge is to design and develop a simple cloud-based College Event Registration System using Zoho Creator. The system should allow students to register for events through an online

form, store the information in the cloud, organize registrations by event and participation category, and provide organizers with reports and a dashboard for monitoring and analysis.

The assignment requires students to apply cloud computing concepts, particularly SaaS, cloud data storage, cloud application development, reporting and data visualization, to a practical college-event-management problem.

C. Problem Statement

Design and develop a simple cloud-based College Event Registration System using Zoho Creator to demonstrate Software as a Service (SaaS). The system should allow students to register for college events by providing their basic details, selecting an event and choosing their participation category. The registration data should be stored and managed through the cloud.

The system should generate simple reports showing the number of registrations for each event and participation category, and the results should be analyzed to identify the most preferred events.

D. Requirements and Constraints

1. Functional Requirements

- The system shall provide a digital registration form for college events.
- The form shall collect the Student Name.
- The form shall collect the Register Number.
- The student shall select the Department.
- The student shall select the Year of study.
- The student shall select an Event Name.
- The system shall allow selection of Participation Type.
- The form shall collect student contact details.
- The system shall record the Registration Date.
- The system shall maintain the Registration Status.

- The system shall generate department-wise participation reports.
- The system shall identify the event with the highest registration count.
- The system shall provide simple visual reports or charts. Identify and highlight the most preferred event(s) based on registration counts.

2. Non-Functional Requirements

- Usability: The interface should be simple and easy for students to understand.
- Reliability: Submitted registration information should be stored correctly without loss.
- Accessibility: The application should be accessible through internet-connected devices.
- Security: Registration and student information should be available only to authorized users.
- Scalability: The system should support an increasing number of event registrations.
- Maintainability: Organizers should be able to manage registration records without rebuilding the application.

3. Constraints

- The system depends on internet connectivity and cloud-platform availability.
- Features may depend on the selected Zoho Creator plan.
- Student registration information should be protected from unauthorized access.
- The project should remain simple enough for academic demonstration.
- The system should use realistic sample data for testing and dashboard validation.

E. Student Work

1. Problem Understanding and Formulation

Problem

Event coordinators need to know how many students are participating, which events are receiving more interest and how participation is distributed among departments. When registration information is maintained manually, preparing these statistics requires additional time and effort.

The main problem is therefore to create a centralized cloud system that can collect registration details and convert the stored information into useful reports.

Existing Process

Student → Manual Registration → Event Organizer → Individual Records → Manual Counting → Event Statistics

Proposed Process

Student → Online Registration → Zoho Creator Application → Cloud Data Storage → Reports and Charts → Registration Analysis

Expected Outcomes

1. Students can register for events using an online form.
2. All registration information is maintained in a centralized location.
3. Event organizers can access registration records more easily.
4. The system provides the total number of registrations.
5. Event-wise registration statistics can be generated.
6. Department-wise participation can be analyzed.
7. Participation Type can be categorized.
8. The most popular event can be identified.
9. The project demonstrates a practical cloud-based application.

Sample Data Fields

Field	Example
Student Name / ID	Nandini / 23AM045
Department	AI&ML
Event Name	Coding Contest
Participation Category	Individual
Email ID	Nandini@example.com
Contact Number	9786543210
Date of Registration	02-09-2026

2. Application of Course Knowledge

Software as a Service (SaaS)

The event registration system is delivered as an online application that can be used through the internet. Zoho Creator provides the required software services for developing and operating the application. Users can access the application without installing separate registration management software.

Cloud-Based Data Storage

The registration records are stored on a cloud platform instead of being maintained only on a single local computer. This enables centralized management of the collected information and supports easier access to authorized users.

Online Application Development

The project applies cloud application development concepts by using Zoho Creator to design forms, data fields, reports and dashboards. The development process does not require the project to manage physical servers.

Data Reporting and Visualization

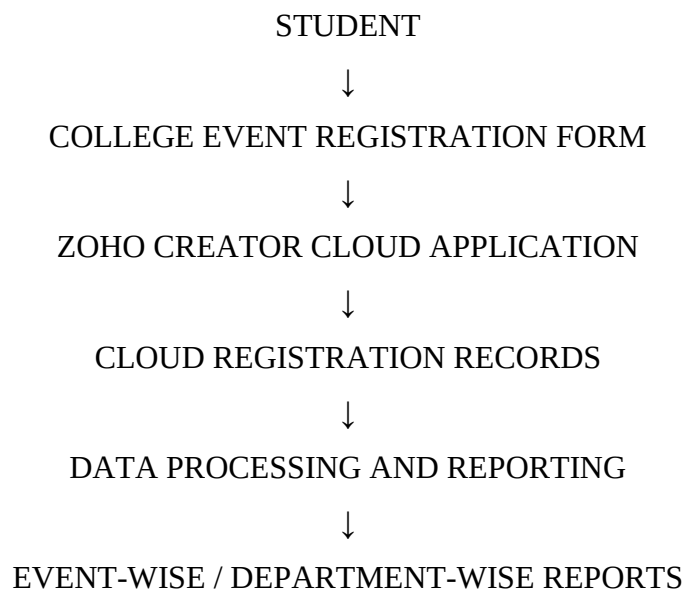
The stored registration information is used to create summaries and visual reports. By grouping records according to event, department and participation type, the system can present useful information for event organizers.

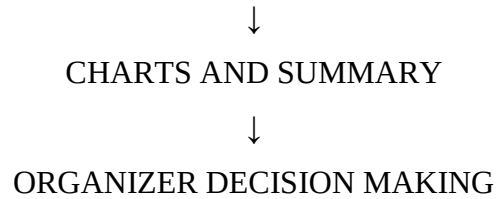
3. Solution / Design / Methodology

Major Modules

Module	Description
Module 1 – Registration Form	Collects student and event registration information.
Module 2 – Registration Database	Maintains submitted records in the cloud.
Module 3 – Report Management	Displays registration details in organized reports.
Module 4 – Data Analysis	Calculates registration counts and identifies popular events

System Architecture





Registration Status Workflow

Submitted → Under Review → Confirmed → Completed

↘ Cancelled

4. Use of Modern Tools

Zoho Creator will be used as the primary cloud application development platform. The project will use its form, report, dashboard and data-management capabilities to create the proposed system.

- Design the event registration form.
- Add fields for student details.
- Configure event selection options.
- Configure participation categories.
- Store submitted records in the cloud.
- Create a complete registration report.
- Create filtered reports for events.
- Create department-based participation reports.
- Display registration totals.
- Create a simple chart or dashboard.

- Analyze the registration results. Evidence of tool usage should be included in the submission through screenshots of the application form, submitted records, registration report and dashboard.

5. Results and Validation

The system can be tested by entering representative registration records and checking whether the submitted information is stored and displayed correctly.

Sample Test Dataset

ID	Student Name	Event	Category	Status
R001	Priya	Tech Symposium	Individual	Confirmed
R002	Arjun	Hackathon	Team	Confirmed
R003	Sneha	Cultural Night	Individual	Confirmed
R004	Karthik	Sports Meet	Team	Confirmed
R005	Divya	Workshop on AI	Individual	Confirmed
R006	Rahul	Tech Symposium	Team	Confirmed
R007	Meena	Coding contest	Individual	Confirmed
R008	Vignesh	Hackathon	Team	Confirmed
R009	Anitha	Sports Meet	Individual	Confirmed
R010	Suresh	Workshop on AI	Team	Confirmed

Dashboard Example

Category	Number
Tech Symposium registrations	8
Hackathon registrations	6
Coding contest	5
Sports Meet registrations	4
Workshop on AI registrations	6
Total registrations	29
Individual category registrations	15
Team category registrations	14

Most preferred event	Coding contest
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Testing and Validation

Test Case	Expected Result
Submit valid registration	Registration should be stored successfully.
Select Tech Symposium event	Registration should be linked to Tech Symposium.
Select Individual category	Registration should display Individual category.
Select Team category	Registration should display Team category.
View registration report	Submitted registration should appear in the report.
Filter by event	Relevant event registrations should be displayed.
Open dashboard	Dashboard should show updated registration statistics.

6. Analysis and Engineering Decision

The proposed cloud-based system is preferable to a manual registration register because it centralizes registration information and provides easier tracking and analysis of event popularity.

Feature	Manual System	Cloud-Based System
Registration submission	Paper/in-person	Online
Data storage	Register/files	Cloud
Tracking	Difficult	Easy
Searching	Manual	Fast filtering/search
Statistics	Manual calculation	Dashboard
Accessibility	Limited	Internet-based
Event popularity analysis	Difficult	Automated via dashboard

Zoho Creator is selected because it provides a suitable cloud environment for developing a small application with forms, data management, reports and dashboards. This makes it appropriate for demonstrating cloud application development and SaaS concepts at an academic level.

7. Broader Considerations

- Sustainability: Digital event registration reduces paper usage.
- Society: The system provides a more organized communication channel between students and event organizers.
- Accessibility: Students can register for events without physically visiting an administrative office.
- Security and Privacy: Student and registration information should be accessible only to authorized users.
- Economics: Centralized digital management can reduce manual administrative effort and paper-related costs.
- Professional Responsibility: Registration records should be maintained accurately and should not be modified or disclosed without authorization.

8. Conclusion

The Cloud-Based College Event Registration System demonstrates how cloud computing can be applied to simplify a common college activity. The application provides a digital method for collecting student registration information and maintaining it through a centralized cloud platform.

The system allows organizers to generate total registration counts, compare registrations between events and examine participation from different departments. The reports and charts make it easier to understand participation patterns and identify the event that attracts the greatest number of students.

The project demonstrates the practical use of SaaS and cloud-based data management. It can be further developed into a larger event management system by adding attendance tracking, notifications and advanced registration controls.

9. References

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2. Erl, T., & Barceló Monroy, E. (2025). *Cloud Computing: Concepts, Technology, Security & Architecture* (2nd ed.). Pearson.
3. National Institute of Standards and Technology (NIST). (2026). *The NIST Definition of Cloud Computing (NIST SP 800-145)*. National Institute of Standards and Technology.
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5. Zoho Corporation. (2026). *Zoho Creator Developer Documentation and Platform Specifications*. Zoho Creator.
6. Zoho Corporation. (2026). *Zoho Creator: Cloud-Based Low-Code Application Development Platform*. Zoho.
7. Agal, S. (2025). *Cloud Computing: Concepts, Technologies, and Applications*. Notion Press.

10. Student Reflection

This assignment helped me understand how cloud technology can be used to solve a practical information management problem. I learned that a SaaS platform can be used to build an application without developing and maintaining a complete physical computing infrastructure.

I gained experience in organizing application data through registration fields and understanding how stored information can be transformed into reports and charts. I also understood the importance of data accuracy and centralized cloud storage.

The project improved my understanding of cloud-based application development and demonstrated how reporting tools can support better decision-making. In the future, the system could include automatic confirmation messages, separate organizer access, attendance recording, registration limits and more detailed analytics.